

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** Bromobenzene

|                                |                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>AJA98</b>                                                                                         |
| <b>Address</b>                 | ThermoFisher Scientific Australia Pty Ltd<br>5 Caribbean Drive, Scoresby<br>VICTORIA 3179, Australia |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>03 9757 4559 or +613 9757 4559</b>                                            |
| <b>Telephone / Fax Numbers</b> | Tel: 1300 735 292<br>Fax: 1800 067 639                                                               |
| <b>E-mail address</b>          | ANZinfo@thermofisher.com                                                                             |

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product contains one or more substance(s) on the Illicit Drug Precursors/Reagents list. Verify requirements related to using, handling and storing these substances. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

Flammable liquids

Category 3

#### Health hazards

Acute Oral Toxicity  
Skin Corrosion/Irritation  
Serious Eye Damage/Eye Irritation  
Specific target organ toxicity - (single exposure)

Category 4  
Category 2  
Category 2  
Category 3

#### Environmental hazards

Chronic aquatic toxicity

Category 2

### Label Elements



Flame



Exclamation Mark

Exclamation Mark



Environment

**Signal Word****Danger****Hazard Statements**

H315 - Causes skin irritation  
 H319 - Causes serious eye irritation  
 H335 - May cause respiratory irritation  
 H302 - Harmful if swallowed  
 H226 - Flammable liquid and vapor  
 H411 - Toxic to aquatic life with long lasting effects

**Precautionary Statements**

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
 P264 - Wash face, hands and any exposed skin thoroughly after handling  
 P271 - Use only outdoors or in a well-ventilated area  
 P280 - Wear protective gloves/protective clothing/eye protection/face protection  
 P270 - Do not eat, drink or smoke when using this product  
 P280 - Wear eye protection/ face protection  
 P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
 P233 - Keep container tightly closed  
 P240 - Ground and bond container and receiving equipment  
 P242 - Use non-sparking tools  
 P243 - Take action to prevent static discharges  
 P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
 P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
 P312 - Call a POISON CENTER or doctor if you feel unwell  
 P362 - Take off contaminated clothing  
 P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell  
 P330 - Rinse mouth  
 P337 + P313 - If eye irritation persists: Get medical advice/attention  
 P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
 P332 + P313 - If skin irritation occurs: Get medical advice/attention  
 P363 - Wash contaminated clothing before reuse  
 P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
 P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
 P403 + P235 - Store in a well-ventilated place. Keep cool  
 P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

Toxicity to Soil Dwelling Organisms  
 Toxic to terrestrial vertebrates  
 This product does not contain any known or suspected endocrine disruptors

## Section 3 - Composition and Information on Ingredients

| Component    | CAS No   | Weight % |
|--------------|----------|----------|
| Bromobenzene | 108-86-1 | 100      |

## Section 4 - First Aid Measures

|                                            |                                                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                          | Remove to fresh air.                                                                                                                             |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water.                                                                                     |
| <b>Skin Contact</b>                        | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes.                                            |
| <b>Eye Contact</b>                         | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician.                             |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                             |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting                                     |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                           |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

No information available.

### Specific Hazards Arising from the Chemical

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Emergency procedures

Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental Precautions

Do not flush into surface water or sanitary sewer system.

### Methods for Containment and Clean Up

#### Clean-up methods - small spillage

Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

#### Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

**Conditions for Safe Storage, Including any Incompatibilities**

Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

The product does not contain any hazardous materials with occupational exposure limits established.

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Exposure Controls****Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time                  | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|------------------------------------|-----------------|-----------------|-----------------------|
| Nitrile rubber | See manufacturer's recommendations | -               | AS/NZS 2161     | (minimum requirement) |
| Neoprene       |                                    |                 |                 |                       |
| Natural rubber |                                    |                 |                 |                       |
| PVC            |                                    |                 |                 |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Long sleeved clothing

**Respiratory Protection**

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:**

Particulates filter conforming to EN 143 (or AUS/NZ equivalent)

|                                        |                                                                                                                     |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Recommended half mask:-</b>         | Particle filtering: EN149:2001 (or AUS/NZ equivalent)<br>When RPE is used a face piece Fit Test should be conducted |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                              |
| <b>Environmental exposure controls</b> | Prevent product from entering drains. Do not allow material to contaminate ground water system.                     |

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                                            |                                          |
|------------------------------------------------|--------------------------------------------|------------------------------------------|
| <b>Appearance</b>                              | Clear                                      |                                          |
| <b>Physical State</b>                          | Liquid                                     |                                          |
| <b>Odor</b>                                    | No information available                   |                                          |
| <b>Odor Threshold</b>                          | No data available                          |                                          |
| <b>pH</b>                                      | Not applicable                             |                                          |
| <b>Melting Point/Range</b>                     | -30.7 °C / -23.3 °F                        |                                          |
| <b>Softening Point</b>                         | No data available                          |                                          |
| <b>Boiling Point/Range</b>                     | Not applicable 155 - 157 °C / 311 314.6 °F |                                          |
| <b>Flash Point</b>                             | Not applicable 55 °C / 131 °F              | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | Not applicable                             | Solid                                    |
| <b>Flammability (solid,gas)</b>                | Not applicable                             | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available                          |                                          |
| <b>Vapor Pressure</b>                          | No data available                          |                                          |
| <b>Vapor Density</b>                           | Not applicable                             | Solid                                    |
| <b>Specific Gravity / Density</b>              | No data available                          |                                          |
| <b>Bulk Density</b>                            | Not applicable                             | Liquid                                   |
| <b>Water Solubility</b>                        | No information available                   |                                          |
| <b>Solubility in other solvents</b>            | No information available                   |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                            |                                          |
| <b>Component</b>                               | <b>log Pow</b>                             |                                          |
| Bromobenzene                                   | 3.12                                       |                                          |
|                                                | 3.14                                       |                                          |
|                                                | 3.25                                       |                                          |
|                                                | 3.29                                       |                                          |
| <b>Autoignition Temperature</b>                | No data available                          |                                          |
| <b>Decomposition Temperature</b>               | No data available                          |                                          |
| <b>Viscosity</b>                               | Not applicable                             | Solid                                    |
| <b>Explosive Properties</b>                    |                                            | explosive air/vapour mixtures possible   |
| <b>Oxidizing Properties</b>                    | No information available                   |                                          |
| <b>Other information</b>                       |                                            |                                          |
| <b>Molecular Formula</b>                       | C <sub>2</sub> H <sub>5</sub> Br           |                                          |
| <b>Molecular Weight</b>                        | 157.01                                     |                                          |

## Section 10 - Stability and Reactivity

|                               |                                                                   |
|-------------------------------|-------------------------------------------------------------------|
| <b>Reactivity</b>             | None known, based on information available                        |
| <b>Stability</b>              | Stable under normal conditions.                                   |
| <b>Conditions to Avoid</b>    | Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Incompatible Materials</b> | None known.                                                       |

**Hazardous Decomposition Products** None under normal use conditions.

**Hazardous Polymerization** No information available.

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

**(a) acute toxicity;**

|            |                   |
|------------|-------------------|
| Oral       | Category 4        |
| Dermal     | No data available |
| Inhalation | No data available |

| Component    | LD50 Oral  | LD50 Dermal | LC50 Inhalation |
|--------------|------------|-------------|-----------------|
| Bromobenzene | 2383 mg/kg |             | 20411 mg/l      |

**(b) skin corrosion/irritation;** Category 2

**(c) serious eye damage/irritation;** Category 2

**(d) respiratory or skin sensitization;**

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

**(e) germ cell mutagenicity;** No data available

**(f) carcinogenicity;** No data available

There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;** No data available

**(h) STOT-single exposure;** Category 3

**Results / Target organs** Respiratory system

**(i) STOT-repeated exposure;** No data available

**Target Organs** No information available.

**(j) aspiration hazard;** Not applicable  
Solid

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

## Section 12 - Ecological Information

#### Ecotoxicity effects

Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component    | Freshwater Fish         | Water Flea       | Freshwater Algae | Microtox                |
|--------------|-------------------------|------------------|------------------|-------------------------|
| Bromobenzene | 5.6 mg/l fathead minnow | 5.8 mg/l Daphnia |                  | EC50 = 9.46 mg/L 30 min |

**Persistence and Degradability** No information available

|                                              |                                                                                                                 |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Persistence</b>                           | Persistence is unlikely.                                                                                        |
| <b>Degradation in sewage treatment plant</b> | Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants. |
| <b>Bioaccumulative Potential</b>             | Bioaccumulation is unlikely                                                                                     |

| Component    | log Pow                      | Bioconcentration factor (BCF) |
|--------------|------------------------------|-------------------------------|
| Bromobenzene | 3.12<br>3.14<br>3.25<br>3.29 | No data available             |

|                                        |                                                                           |
|----------------------------------------|---------------------------------------------------------------------------|
| <b>Mobility</b>                        | No information available.                                                 |
| <b>Endocrine Disruptor Information</b> | This product does not contain any known or suspected endocrine disruptors |
| <b>Persistent Organic Pollutant</b>    | This product does not contain any known or suspected substance            |
| <b>Ozone Depletion Potential</b>       | This product does not contain any known or suspected substance            |

## Section 13 - Disposal Considerations

|                                            |                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations. |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                               |                                                                                                                                                                                                                                          |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Contaminated Packaging</b> | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition. |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                          |                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Other Information</b> | Chemical wastes should be disposed through a licensed commercial waste collection service. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not let this chemical enter the environment. Do not empty into drains. |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Section 14 - Transport Information

### IMDG/IMO

|                                |              |
|--------------------------------|--------------|
| <b>UN-No</b>                   | UN2514       |
| <b>Proper Shipping Name</b>    | BROMOBENZENE |
| <b>Hazard Class</b>            | 3            |
| <b>Subsidiary Hazard Class</b> | P            |
| <b>Packing Group</b>           | III          |

| Component                        | IMDG Marine Pollutant                    |
|----------------------------------|------------------------------------------|
| Bromobenzene<br>108-86-1 ( 100 ) | IMDG regulated marine pollutant (UN2514) |

### ADG

|                             |              |
|-----------------------------|--------------|
| <b>UN-No</b>                | UN2514       |
| <b>Proper Shipping Name</b> | BROMOBENZENE |
| <b>Hazard Class</b>         | 3            |
| <b>Packing Group</b>        | III          |

| Component                        | Hazchem Code |
|----------------------------------|--------------|
| Bromobenzene<br>108-86-1 ( 100 ) | 2Y           |

### IATA

|                             |              |
|-----------------------------|--------------|
| <b>UN-No</b>                | UN2514       |
| <b>Proper Shipping Name</b> | BROMOBENZENE |
| <b>Hazard Class</b>         | 3            |

|                        |                                                                                                          |
|------------------------|----------------------------------------------------------------------------------------------------------|
| Packing Group          | III                                                                                                      |
| Environmental hazards  | Dangerous for the environment<br>Product is a marine pollutant according to the criteria set by IMDG/IMO |
| Special Precautions    | No special precautions required                                                                          |
| Additional information | None known                                                                                               |

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations Australia

See section 8 for national exposure control parameters.

#### **Standard for the Uniform Scheduling of Medicines and Poisons**

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

#### **Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component               | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|-------------------------|-------------------------------------------------------------|------------------------|
| Bromobenzene - 108-86-1 | Present                                                     | -                      |

#### **Australian - Illicit Drug Precursors/Reagents Substance List**

This product contains one or more substance(s) on the Illicit Drug Precursors/Reagents list. Verify requirements related to using, handling and storing these substances.

#### **Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

| Component               | Australian - Illicit Drug Precursors/Reagents Substance List | Chemicals of Security Concern |
|-------------------------|--------------------------------------------------------------|-------------------------------|
| Bromobenzene - 108-86-1 | Category 1                                                   |                               |

#### **Legend**

Category 1 - Chemicals that require an End User Declaration with each purchase and may only be sold to 'account customers' or customers that are prepared to open an account. Supply of these chemicals to End Users or Distributors must be delayed for a period of not less than 24 hours

#### **National pollutant inventory** Not applicable

#### **Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

#### **International Inventories**

| Component    | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|--------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Bromobenzene | X    | X     | 203-623-8 | -      | X    | X   | -    | X     | X    | X    | X     | KE-03624 |



**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**MARPOL** - International Convention for the Prevention of Pollution from Ships

| Component               | IMDG Marine Pollutant                    |
|-------------------------|------------------------------------------|
| Bromobenzene - 108-86-1 | IMDG regulated marine pollutant (UN2514) |

### **Basel convention on the control of transboundary movements of hazardous wastes and their disposal**

Take note that wastes may be subject to export, import, or transit controls pursuant to the Basel convention and/or local regulations implementing the Basel convention.

| Component               | Basel Convention (Hazardous Waste) | Australian Hazardous Waste Act - Categories of Wastes to Be Controlled |
|-------------------------|------------------------------------|------------------------------------------------------------------------|
| Bromobenzene - 108-86-1 | Annex I - Y45                      | Y45 except substances referenced in Annex I                            |

| Component    | CAS No   | OECD HPV       | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|--------------|----------|----------------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Bromobenzene | 108-86-1 | Not applicable | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

### **Authorisation/Restrictions according to EU REACH**

| Component    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Bromobenzene | -                                                                   | Use restricted. See entry 75. (see link for restriction details)              | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

## **Section 16 - Other Information**

### Legend

**AICS** - Australian Inventory of Chemical Substances  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List  
**IECSC** - Chinese Inventory of Existing Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**TWA** - Time Weighted Average  
**IARC** - International Agency for Research on Cancer  
**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from

**NZIoC** - New Zealand Inventory of Chemicals  
**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
**ENCS** - Japanese Existing and New Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances  
**CAS** - Chemical Abstracts Service  
**ACGIH** - American Conference of Governmental Industrial Hygienists  
**PNEC** - Predicted No Effect Concentration  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

## Ships

**NZS 5433:2020** - Transport of Dangerous Goods on Land**LD50** - Lethal Dose 50%**EC50** - Effective Concentration 50%**WEL** - Workplace Exposure Limit**DNEL** - Derived No Effect Level**POW** - Partition coefficient Octanol:Water**vPvB** - very Persistent, very Bioaccumulative**VOC** - (Volatile Organic Compound)**ADG** - Australian Code for the Transport of Dangerous Goods by Road and Rail**OECD** - Organisation for Economic Co-operation and Development**LC50** - Lethal Concentration 50%**ATE** - Acute Toxicity Estimate**RPE** - Respiratory Protective Equipment**NOEC** - No Observed Effect Concentration**BCF** - Bioconcentration factor**PBT** - Persistent, Bioaccumulative, Toxic**Key literature references and sources for data**<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Revision Date**

12-Mar-2025

**Revision Summary**

Update to GHS format.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet